

# BRENDAN LO

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## EDUCATION

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|-------------------------------------------------------------------------|-----------------------------------------|
| <b>University of Chicago</b><br>B.S., Computer Science and Mathematics  | <i>Expected May 2030</i><br>Chicago, IL |
| <b>Great Neck South High School</b><br>GPA 96/100 unweighted · SAT 1540 | <i>June 2026</i><br>Great Neck, NY      |

## RESEARCH EXPERIENCE

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| <b>Software Engineer Intern, Shasha Lab</b> · New York University<br>• Built an AI calorie-tracking app beating commercial-app accuracy with Bayesian MAP estimation, LiDAR depth capture, ARKit, and the Gemini API in Swift.                                            | <i>Summer 2026</i> |
| <b>Data Analysis Intern, Kuhn &amp; Liu Lab</b> · Weill Cornell Medical College<br>• Characterized calcium's role in cardiomyocyte proliferation via computational image and signal analysis in Python and Fiji/ImageJ; published in the <i>Journal of Cell Biology</i> . | <i>Summer 2025</i> |
| <b>Data Analysis Intern (Genomics), Long Lab</b> · Weill Cornell Medical College<br>• Applied RNA-seq analysis, PCR, Western blot, and cloning in R and Python to identify JARID2 gene targets and probe PRC2 in congenital heart disease.                                | <i>Summer 2024</i> |

## ENGINEERING & LEADERSHIP

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| <b>Co-Founder &amp; Co-President, Web Development Club</b><br>• Led full-stack teams across 70+ members in TypeScript, React, and Supabase; shipped and maintained 20+ sites for nonprofits, businesses, and school organizations. | <i>Oct 2024 – June 2026</i> |
| <b>Chief Technology Officer, Kudos Connect</b><br>• Directed a 15-person team building the site and newsletter, reaching 7,000+ students.                                                                                          | <i>Jan 2024 – Present</i>   |
| <b>Head Webmaster &amp; Lead Programmer, FIRST Robotics Team 2638</b><br>• Wrote trajectory-generation algorithms and robot control code; team advanced to the FRC World Championship (2025).                                      | <i>Sep 2024 – June 2026</i> |
| <b>Software &amp; App Lead, Programming Club</b><br>• Taught 20+ students Python, Java, and applied ML; led 10 students building a Swift school app serving 1,300+ students.                                                       | <i>Jan 2024 – June 2026</i> |

## PROJECTS

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| <b>SIDEWALK</b> — 2nd Place, NYC Hackathon (Shopify × Base44)<br>• Turned NYC's PDF-only vending-restriction guide into a queryable geospatial dataset (Deno/TypeScript ETL, street-centerline BFS resolution) behind an offline-first React/Leaflet PWA in seven languages, covered by 180+ unit and Playwright E2E tests.                                                                                     | <i>2026</i>           |
| <b>SATsaurus</b> — Co-Founder & Software Lead<br>• Led a team of five shipping a gamified SAT-prep web app in Next.js and Supabase, reaching 1,000+ students.                                                                                                                                                                                                                                                   | <i>2025 – 2026</i>    |
| <b>ML Drug-Discovery Pipeline</b> — Lead Developer<br>• Ran Random Forest virtual screening and GROMACS molecular-dynamics simulations to identify PRC2-targeting drug candidates.<br>• Extended the work into three co-authored workshop posters at the International Conference on Machine Learning (ICML) 2026 on drug-binding prediction, retrovirus reactivation, and biological-forecasting scaling laws. | <i>2024 – 2025</i>    |
| <b>Force Network</b> — Co-Founder & Software Lead<br>• Built and operate a Linux hosting hub for 100+ game servers, delivering Docker deployment tooling and bandwidth to 800+ users.                                                                                                                                                                                                                           | <i>2022 – Present</i> |

## PUBLICATIONS & PRESENTATIONS

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| Liu, H., Ammanamanchi, N., Mich-Basso, J., Panama, B., Li, Y., Huang, W., Almeida, D., Lewarchik, C., Lo, B., Wu, Y., Gotthardt, M., Kotlikoff, M., Baehr, W., Rasmusson, R., Salama, G., & Kuhn, B. (2026). Sequential changes in calcium transients during M phase regulate cardiomyocyte proliferation. <i>Journal of Cell Biology</i> . doi:10.1083/jcb.202505134 |
| Cheng, B., Jin, A., Zhang, J., Pemmaraju, A., Lo, B., & Chang, J. (2026). A horizon-dependent intrinsic-dimension theory of scaling for biological forecasting. <i>HiLD Workshop, ICML 2026</i> (poster).                                                                                                                                                             |
| Cheng, B., Jin, A., Zhang, J., Lo, B., Pemmaraju, A., & Chang, J. (2026). ERVNet: a three-module framework for predicting endogenous retrovirus reactivation, gene propagation, and immunogenicity. <i>FM4LS Workshop, ICML 2026</i> (poster).                                                                                                                        |
| Cheng, B., Jin, A., Zhang, J., Pemmaraju, A., Chang, J., & Lo, B. (2026). PROTEUS: predicting how post-translational modifications alter drug binding affinity. <i>FM4LS Workshop, ICML 2026</i> (poster).                                                                                                                                                            |

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, C#, TypeScript, JavaScript, Swift, SQL, R, Bash  
**Infrastructure:** Linux, Docker, AWS, Google Cloud Platform, nginx, Git/GitHub, CI/CD (GitHub Actions), Playwright, Vercel  
**Web & mobile:** React, Next.js, Node.js, Deno, Express, REST APIs, PostgreSQL, Supabase, MongoDB, SwiftUI, ARKit, Gemini API  
**ML & scientific:** PyTorch, scikit-learn, NumPy, pandas, Bayesian MAP estimation, RNA-seq analysis, molecular docking, GROMACS, Schrödinger Maestro, Fiji/ImageJ

## AWARDS & HONORS

Y Combinator Startup School 2026, selected attendee (\$25,000+ in API credits) · Eagle Scout · Science Olympiad National Qualifier and 2nd Place, New York State · Engineering Inspiration Award, FIRST Robotics World Championship · Honorable Mention, Long Island Science & Engineering Fair